# AWS Certified ML Engineer Associate (MLA-C01)

## SECTION 3 — Data Transformation, Integrity & Feature Engineering

**PART 2 STUDY NOTES**

AWS Managed ML Services + Data Quality Tools

SageMaker Deep Dive • AWS Glue • Amazon Athena

### ■ Topics Covered in This Part

1. SageMaker Data Processing Deep Dive
2. SageMaker Ground Truth & Labeling
3. SageMaker Data Wrangler
4. SageMaker Feature Store
5. SageMaker Model Monitor & Clarify
6. AWS Glue — ETL & Data Quality
7. AWS Glue DataBrew
8. Amazon Athena — Advanced Features

# 1. SageMaker — Data Processing Deep Dive

## 1.1 The SageMaker ML Workflow

SageMaker is AWS's fully managed ML platform that handles the **entire machine learning lifecycle** — from data ingestion all the way to production deployment.

| Phase | What Happens | Key Services Used |
|---|---|---|
| Data Prep | Fetch, clean, transform, engineer features | S3, Glue, Data Wrangler, Processing Jobs |
| Training | Train model on compute instances | SageMaker Training, ECR, S3 |
| Evaluation | Measure accuracy, bias, explainability | Clarify, Model Monitor |
| Deployment | Serve predictions at scale | Endpoints, Batch Transform, Neo |

## 1.2 SageMaker Training Architecture

Understanding the architecture is crucial for exam questions about data flow and security:

- Training data must live in **Amazon S3** — SageMaker pulls from S3 to training instances
- Training code is packaged as a **Docker image stored in Amazon ECR**
- Model artifacts (weights, params) are written back to S3 after training
- An **Inference code image** (also in ECR) is used at deployment time
- The **Endpoint** serves the model to client applications in real-time

> ■ **EXAM TIP**
> If an exam question asks 'where does SageMaker store training data / model output?' — the answer is almost always AMAZON S3. Training code lives in ECR. These two facts appear frequently!

## 1.3 SageMaker Notebooks & Data Access

- Notebook Instances are **EC2 instances** managed by SageMaker
- Pre-loaded with scikit-learn, Spark, TensorFlow, and many other frameworks
- Can access S3 data directly and spin up training instances
- Can deploy trained models from within the notebook

### Ideal Data Formats for SageMaker:

| Algorithm Type | Preferred Format | Why |
|---|---|---|
| Most built-in algos | RecordIO / Protobuf | Optimized for speed, pipe mode compatible |
| XGBoost | CSV or LibSVM | Native format for XGBoost |
| NLP / Text | JSON Lines | One JSON object per line |
| Images | RecordIO or raw | Pipe mode for large datasets |

**Pipe Mode vs File Mode:**

- **File Mode: S3 data copied to training instance disk — simple but slower for large data**
- **Pipe Mode: Data streamed directly from S3 — faster, no local disk needed, ideal for large datasets with RecordIO**

### Data Sources Beyond S3:

- Amazon Athena — query S3 data lake with SQL before training
- Amazon EMR — process big data with Spark, then feed to SageMaker
- Amazon Redshift — pull structured data from data warehouse
- Amazon Keyspaces — AWS's managed Cassandra (NoSQL)
- Apache Spark integration — run feature engineering at scale, output to SageMaker

## 1.4 SageMaker Processing Jobs

SageMaker Processing is a dedicated feature for running data pre/post-processing at scale, separate from training jobs.

### How Processing Jobs Work:

- Step 1: Copy input data from S3 to a managed processing container
- Step 2: Run your processing script inside the container
- Step 3: Output processed data back to S3

### Container Options:

- **SageMaker Built-in containers: scikit-learn, Spark, XGBoost pre-built**
- **Custom containers: Bring your own Docker image for any framework**

**■ KEY POINTS**

- Processing Jobs are SEPARATE from training — use them for feature engineering, data validation, model evaluation
- Input/output always goes through S3
- Can run scikit-learn, PySpark, or custom scripts
- Useful for post-training model evaluation (compute metrics after training)

## 1.5 SageMaker Training Job Options

SageMaker supports a wide range of training options — knowing which to use when is exam-critical:

| Training Option | Description | When to Use |
|---|---|---|
| Built-in Algorithms | AWS-optimized: XGBoost, Linear Learner, K-Means, etc. | Standard ML tasks, optimized performance |
| Spark MLLib | Distributed ML with Apache Spark | Already using Spark ecosystem |
| Custom Python (TF/PyTorch) | Your own training script | Custom architectures, research |
| Scikit-Learn / XGBoost | Framework containers with your script | Standard models with custom logic |

| Training Option | Description | When to Use |
|---|---|---|
| Hugging Face | Pre-built NLP containers | NLP, transformers, LLMs |
| Custom Docker Image | Full control over environment | Exotic dependencies or frameworks |
| AWS Marketplace | Third-party algorithms | Specialized commercial algorithms |

## 1.6 SageMaker AI Domains & VPC

**SageMaker Domains** are organizational containers that group users, apps, and resources:

- Share an **EFS (Elastic File System)** volume
- **User profiles own personal apps (private EFS directory)**
- **Shared spaces enable collaboration (communal IDE, shared EFS directory)**

### VPC Configuration for Domains:

- By default: 2 VPCs — one managed by SageMaker (internet), one your own (encrypted to EFS)
- **VPC Only mode: all traffic routed through your VPC (select at domain creation)**
- You specify VPC, subnets, and security groups for your VPC

> ■ **EXAM TIP**
>
> For security-focused exam questions: 'VPC Only' mode in SageMaker Domains means NO internet access — all traffic goes through your private VPC. This is the high-security option.

## 1.7 Deploying Trained Models

| Deployment Type | Use Case | Key Feature |
|---|---|---|
| Persistent Endpoint | Real-time individual predictions on demand | Low latency, always-on |
| Batch Transform | Predictions for entire dataset at once | Cost-effective for bulk inference |
| Inference Pipelines | Multiple containers chained together | Pre-process → Model → Post-process |
| SageMaker Neo | Deploy to edge devices (IoT, mobile) | Compiles model for target hardware |
| Elastic Inference | Accelerate deep learning models | Add GPU without full GPU instance |
| Shadow Testing | Test new model alongside current | Catch errors before full rollout |

- **Automatic Scaling: Increase number of endpoint instances based on traffic load**

> ■ **EXAM TIP**
> Exam loves asking about deployment types: Real-time single prediction = Persistent Endpoint. Entire dataset at once = Batch Transform. Edge device = SageMaker Neo. Test new model safely = Shadow Testing.

# 2. SageMaker Ground Truth & Labeling

## 1. What is Ground Truth?

Ground Truth solves the **cold-start problem** in supervised ML: you need labeled training data, but none exists yet. It manages human labelers who tag your raw data.

- Typical use case: image classification — humans must tag thousands of images before training
- Ground Truth manages the entire labeling workflow end-to-end

## 2. The Smart Labeling Loop

Ground Truth's key innovation is its **active learning loop** that reduces labeling cost by up to **70%**:

| Step | What Happens |
|---|---|
| 1. Raw data ingested | Images/text/audio uploaded to S3 |
| 2. Humans label initial batch | Establishes ground truth labels |
| 3. Ground Truth trains its own model | Learns from human-labeled examples |
| 4. Model auto-labels easy cases | High-confidence predictions accepted automatically |
| 5. Only ambiguous data goes to humans | Hard cases, low-confidence predictions only |
| 6. Loop continues | Model improves, fewer human labels needed over time |

■ **EXAM TIP**

Key number: Ground Truth's active learning loop reduces labeling cost by UP TO 70%. The model learns which items are 'easy' and labels them automatically — humans only see the 'hard' ones.

## 2.3 Who Are the Labelers?

| Labeler Type | Description | Best For |
|---|---|---|
| Amazon Mechanical Turk | Crowdsourced marketplace workers worldwide | Large volumes, general tasks, low cost |
| Your Internal Team | Your own employees/contractors | Proprietary/confidential data, domain expertise |
| Professional Labeling Companies | Third-party specialized vendors | High quality, complex annotation tasks |

## 2.4 Ground Truth Plus

Ground Truth Plus is a **fully managed, turnkey labeling service** where AWS experts handle everything:

- You fill out an intake form → AWS contacts you to discuss pricing and requirements
- AWS manages the entire labeling workflow and labeler team
- Track progress via the **Ground Truth Plus Project Portal**
- Receive completed labeled data from S3 when the project is done

### Alternatives to Human Labeling:

| Service | What It Labels | How |
|---|---|---|
| Amazon Rekognition | Images | Automatic image classification & object detection |
| Amazon Comprehend | Text | Topic modeling, sentiment analysis, entity recognition |
| Pre-trained models | Any modality | Use unsupervised techniques or existing models |

## 2.5 Amazon Mechanical Turk (MTurk)

MTurk is a **crowdsourcing marketplace** for distributing simple human intelligence tasks to a distributed virtual workforce.

- **How it works: Post tasks (HITs - Human Intelligence Tasks) → Workers complete them → You pay per task**
- **Example: 10,000,000 images × $0.10 per image = $1,000,000 total labeling cost**
- **Use cases: Image classification, data collection, transcription, business processing**
- **Integrates with: Amazon A2I (Augmented AI), SageMaker Ground Truth**

■■ **REMEMBER / WATCH OUT**

- MTurk workers are NOT specialists — for complex/medical/legal data, use professional labeling companies

- Data sent to MTurk workers is NOT private — do not use for confidential/proprietary datasets

- Ground Truth integrates directly with MTurk — no separate setup needed

# 3. SageMaker Data Wrangler

## 3.1 What is Data Wrangler?

Data Wrangler is a **visual, no-code/low-code data preparation tool** inside SageMaker Studio. It provides a GUI for importing, visualizing, transforming, and exporting data for ML.

### Core Workflow:

| Step | Action | Details |
|---|---|---|
| 1. Import | Connect to data source | S3, Athena, Redshift, Feature Store, JDBC, etc. |
| 2. Visualize | Explore distributions, correlations | Histograms, scatter plots, box plots |
| 3. Transform | Apply 300+ built-in transforms | Or custom pandas/PySpark/SQL code |
| 4. Quick Model | Train a quick model to validate features | See feature importance, model metrics |
| 5. Export | Push data flow to production | To S3, Processing Job, Pipeline, Feature Store |

## 3.2 Data Sources for Data Wrangler

| Source | Type |
|---|---|
| Amazon S3 | Object storage — most common |
| Amazon Athena | Serverless SQL on S3 data lake |
| Amazon Redshift | Data warehouse |
| AWS Lake Formation | Governed data lake |
| SageMaker Feature Store | ML feature repository |
| JDBC connections | Databricks, Snowflake, SaaS databases |

## 3.3 Notable Data Wrangler Capabilities

### Image Data Transformations:

- Resize, crop, rotate, flip images
- Enhance contrast, brightness, saturation
- Corrupt images (for augmentation / robustness testing)

### Handling Imbalanced Data:

- **Random Oversampling — duplicate minority class samples**
- **Random Undersampling — remove majority class samples**
- **SMOTE — Synthetic Minority Over-sampling TEchnique (best option)**

### Missing Data Imputation:

- Mean / median / mode replacement
- Custom fill values
- Drop rows/columns with missing values

### Outlier Handling:

- Standard deviation-based detection and removal
- IQR (Interquartile Range) method

### Dimensionality Reduction:

- **PCA (Principal Component Analysis) — reduce many features to fewer principal components**

> ■ **EXAM TIP**
> Data Wrangler is the AWS exam's go-to answer for 'visual/no-code data preparation within SageMaker Studio'. It has 300+ built-in transforms AND supports custom transforms via pandas, PySpark, or PySpark SQL. The 'Quick Model' feature lets you validate your feature engineering quickly.

## 3.4 Data Wrangler Troubleshooting

| Problem | Cause | Solution |
|---|---|---|
| Permission denied to data source | Missing IAM roles | Add AmazonSageMakerFullAccess policy to Studio user |
| Can't access S3 bucket | Missing S3 permissions | Add s3:GetObject, s3:ListBucket to IAM role |
| Instance type not available | EC2 quota limit hit | Request quota increase via Service Quotas → SageMaker → Studio KernelGateway Apps (ml.m5.4xlarge) |

# 4. SageMaker Feature Store

## 1. The Problem Feature Store Solves

In production ML systems, multiple teams often re-compute the same features independently. This wastes compute, introduces inconsistencies, and creates data silos. Feature Store provides a **central, versioned, secure repository** for ML features.

- A **feature** is any property used to train an ML model (age, income, click rate, etc.)
- Features need to be **consistent** between training time and inference time
- Feature Store ensures **training-serving consistency** — a critical ML engineering problem

## 2. Feature Store Architecture

| Component | Description | Access Method |
|---|---|---|
| Feature Group | A collection of related features (like a table) | — |
| Record | A single entity's feature values at a point in time | — |
| Online Store | Low-latency key-value store for real-time inference | PutRecord / GetRecord API |
| Offline Store (S3) | Historical feature data for model training | Athena, Data Wrangler, anything that hits S3 |
| Glue Data Catalog | Auto-created catalog for offline store | Athena SQL queries |

## 4.3 Data Ingestion to Feature Store

Feature Store supports both **streaming** and **batch** ingestion:

| Ingestion Type | Sources | Goes To |
|---|---|---|
| Streaming (real-time) | Amazon Kinesis, Amazon MSK (Kafka), direct API | Online Store first, then Offline Store |
| Batch | Spark, Data Wrangler, processing jobs, S3 | Offline Store (S3) |

### Feature Store Data Structure:

- **Record Identifier — unique key for each entity (e.g., customer_id)**
- **Feature Name — the feature being stored (e.g., age, purchase_count)**
- **Event Time — when the feature value was recorded (enables point-in-time queries)**

**■ KEY POINTS**

- Online Store = fast, real-time, low-latency (inference time)
- Offline Store = historical, batch, S3-backed (training time)
- Glue Data Catalog is automatically created for offline store — query with Athena
- Event time enables point-in-time correctness — no data leakage from future values

## 4.4 Feature Store Security

| Security Feature | Details |
|---|---|
| Encryption at Rest | Both online and offline stores encrypted |
| Encryption in Transit | TLS for all data movement |
| KMS Integration | Works with customer master keys (CMK) |
| IAM Access Control | Fine-grained permissions per feature group |
| AWS PrivateLink | Private connectivity — no traffic over public internet |

**■ EXAM TIP**

For exam questions about 'sharing features across models/teams' or 'consistent features between training and inference' — the answer is SageMaker Feature Store. For real-time inference = Online Store. For training = Offline Store (S3).

# 5. SageMaker Model Monitor & Clarify

## 1. SageMaker Model Monitor Overview

After deployment, models degrade over time as real-world data changes. Model Monitor **continuously monitors deployed models** and alerts you to quality issues via CloudWatch.

### Key Capabilities:

- Detect **data drift** — input data distribution shifts from training baseline
- Detect **model quality drift** — accuracy degrades over time
- Detect **bias drift** — model becomes more biased after deployment
- Detect **feature attribution drift** — which features matter changes over time
- All with **zero code** required!

### Infrastructure:

- Data stored in S3, secured and encrypted
- Monitoring jobs run on a **Monitoring Schedule** (e.g., every hour, daily)
- Metrics emitted to **CloudWatch** for alerting and dashboards
- CloudWatch alarms trigger corrective action (retrain model, audit data)
- Integrates with Tensorboard, QuickSight, Tableau, SageMaker Studio

## 2. Four Types of Model Monitoring

| Monitoring Type | What It Detects | Baseline Needed |
|---|---|---|
| Data Quality Drift | Statistical properties of input features change (mean, std dev, distribution) | Baseline from training data statistics |
| Model Quality Drift | Accuracy, precision, recall, F1 drop over time | Baseline model quality metrics |
| Bias Drift | Model fairness degrades for specific demographic groups | Pre-training bias metrics (via Clarify) |
| Feature Attribution Drift | Relative importance of features changes significantly | NDCG score baseline from training |

**Feature Attribution Drift** uses **NDCG (Normalized Discounted Cumulative Gain) score** — compares feature rankings in training data vs. live production data.

## 5.3 SageMaker Clarify

Clarify is integrated into Model Monitor and provides two key capabilities:

### ① Bias Detection:

- Detects potential bias BEFORE training (pre-training bias metrics)
- Detects bias AFTER training and in production (post-training, via Model Monitor)
- Identifies imbalances across demographic groups: age, gender, income, etc.

### ② Model Explainability:

- Understands WHICH features contribute most to predictions
- Uses **SHAP (SHapley Additive exPlanations)** as an approximation technique
- For Time Series: uses **Asymmetric Shapley Values** — contribution at each time step

## 5.4 Pre-Training Bias Metrics in Clarify

These metrics help you understand bias in your training data BEFORE you spend compute on training:

| Metric | Full Name | What It Measures |
|---|---|---|
| CI | Class Imbalance | One demographic group has fewer training samples |
| DPL | Difference in Proportions of Labels | Imbalance of positive outcomes between groups |
| KL / JS | Kullback-Leibler / Jensen-Shannon Divergence | How much outcome distributions diverge between groups |
| LP | Lp-norm | P-norm difference between outcome distributions |
| TVD | Total Variation Distance | L1-norm difference between distributions |
| KS | Kolmogorov-Smirnov | Maximum divergence between outcome distributions |
| CDD | Conditional Demographic Disparity | Disparity of outcomes by subgroups and overall |

## 5.5 Partial Dependence Plots (PDPs) & Shapley Values

### Partial Dependence Plots:

- Shows how predicted outcome DEPENDS on a specific input feature
- Plot shows: as feature X increases from low to high, how does prediction change?
- Also shows data distributions for each bucket of feature values

### Shapley Values (SHAP):

- Originated in **game theory** — measures each player's (feature's) contribution to the outcome
- In ML: measures impact of **dropping each individual feature** on prediction
- SageMaker Clarify uses **SHAP** as a computationally tractable approximation
- Gets computationally expensive with many features — SHAP approximates this

> ■ **EXAM TIP**
> Exam question pattern: 'How do you explain which features are most important to your model's predictions?' Answer = SageMaker Clarify with SHAP values. 'How do you detect if your model is becoming biased after deployment?' = Model Monitor + Clarify (bias drift monitoring).

# 6. AWS Glue — ETL & Data Quality

## 6.1 What is AWS Glue?

AWS Glue is a **fully managed, serverless ETL (Extract, Transform, Load) service** with a built-in data catalog for schema discovery and management.

| Component | Function |
|---|---|
| Glue Crawler | Scans data sources, infers schema, populates Data Catalog |
| Glue Data Catalog | Central metadata repository — table definitions, schemas |
| Glue ETL Jobs | Apache Spark-based transform scripts (Python/Scala) |
| Glue Studio | Visual drag-and-drop ETL workflow builder |
| Glue Data Quality | Define and enforce data quality rules (DQDL) |
| Glue DataBrew | Visual data preparation tool for analysts |

## 6.2 Glue Crawler & Data Catalog

The Glue Crawler is the entry point for bringing external data into the Glue ecosystem:

- Crawler **scans S3** (or RDS, Redshift, DynamoDB, JDBC databases)
- Infers schema and data types automatically
- Can run periodically (scheduled) to pick up new data
- Populates the **Glue Data Catalog** with table definitions
- Original data stays in S3 — only metadata is stored in the catalog

### Once Cataloged, Data is Queryable By:

- **Amazon Athena — serverless SQL on S3**
- **Amazon Redshift Spectrum — query S3 from Redshift**
- **Amazon EMR — process with Spark/Hive**
- **Amazon QuickSight — BI visualization**

## 6.3 S3 Partitions & Glue Performance

How you organize data in S3 directly impacts query performance. **Design your partition structure based on how you'll query the data.**

| Query Pattern | Optimal S3 Structure | Example |
|---|---|---|
| Primarily by TIME | yyyy/mm/dd/device | Find all sensor readings on 2024-01-15 |
| Primarily by DEVICE | device/yyyy/mm/dd | Find all readings from Device-001 ever |

**■■ REMEMBER / WATCH OUT**

- Glue crawler EXTRACTS partitions based on S3 folder structure — design structure BEFORE loading data

- Wrong partition design = full table scans = slow, expensive queries
- Partition pruning is the #1 way to improve Athena and Glue query performance

## 6.4 AWS Glue Studio

Glue Studio provides a **visual interface for building ETL pipelines** as DAGs (Directed Acyclic Graphs):

- **Sources: S3, Kinesis, Kafka, JDBC databases**
- **Transforms: Filter, join, aggregate, sample, map, custom SQL**
- **Targets: S3 or Glue Data Catalog**
- **Partitioning: Full support for output partitioning**
- **Job Dashboard: Visual overview of job status, run history, and run times**

## 6.5 AWS Glue Data Quality

Glue Data Quality lets you define and enforce **data quality rules** within ETL pipelines:

- Rules can be created **manually** or **automatically recommended** by Glue
- Uses **DQDL (Data Quality Definition Language)** to express rules
- Integrates directly into Glue ETL Jobs — runs as part of the pipeline
- On failure: can **fail the job** (hard stop) or just **report to CloudWatch** (soft warning)

### Example DQDL Rules:

```
IsComplete "column_name" — no null values
ColumnValues "age" between 0 and 120 — valid range
IsPrimaryKey "user_id" — unique, not null
```

**■ EXAM TIP**

Glue vs Athena anti-patterns: Athena is NOT for ETL (use Glue). Glue is NOT for highly formatted reports/visualization (use QuickSight). This comes up in exam scenario questions!

# 7. AWS Glue DataBrew

## 7.1 What is DataBrew?

AWS Glue DataBrew is a **visual data preparation tool** aimed at data analysts and scientists. It requires no coding and offers 250+ ready-made transformations.

| Feature | Details |
|---|---|
| Target Users | Data analysts, data scientists — no engineering required |
| Input Sources | S3, Redshift, Snowflake (via custom SQL), databases |
| Output | Always to S3 |
| Transforms | 250+ ready-made transformations |
| Custom Transforms | Define custom SQL (from Redshift and Snowflake) |
| Recipes | Reusable sequences of transforms saved as jobs in a project |
| Data Quality | Define quality rules within DataBrew profiles |

## 7.2 DataBrew Recipes

A **Recipe** is a named, reusable sequence of transformation steps:

- Created interactively in the DataBrew visual UI
- Saved and reused across multiple datasets or scheduled runs
- Runs as a **Job** within a larger DataBrew **Project**
- Can be exported and applied to new data sources

### Example Recipe Action (JSON format):

```
{"RecipeAction": {"Operation": "NEST_TO_MAP", "Parameters": {"sourceColumns":
"[\"age\",\"weight_kg\"]", "targetColumn": "patientData", "removeSourceColumns": "true"}}}
```

## 7.3 Handling PII in DataBrew

DataBrew has built-in **PII (Personally Identifiable Information) handling** capabilities — important for GDPR, HIPAA compliance:

| Operation Type | DataBrew Function | What It Does |
|---|---|---|
| Substitution | REPLACE_WITH_RANDOM | Replace PII with random values |
| Shuffling | SHUFFLE_ROWS | Randomly reorder rows to break PII linkage |
| Deterministic Encrypt | DETERMINISTIC_ENCRYPT | Same input always gives same output (pseudonymization) |
| Probabilistic Encrypt | ENCRYPT | Random encryption — different output each time |
| Decryption | DECRYPT | Reverse deterministic encryption |
| Deletion / Nulling | DELETE | Remove PII fields entirely |

| Operation Type | DataBrew Function | What It Does |
|---|---|---|
| Masking | MASK_CUSTOM, MASK_DATE, etc. | Partially obscure values (e.g., ***-**-1234) |
| Hashing | CRYPTOGRAPHIC_HASH | One-way transform — SHA-256 etc. |

### How to Enable PII Detection:

- Step 1: Run a **DataBrew profile job** on your dataset
- Step 2: Enable **PII statistics** in the profile job settings
- Step 3: DataBrew highlights columns that likely contain PII
- Step 4: Apply appropriate PII transformation from the list above

## 7.4 DataBrew Security

| Security Layer | Details |
|---|---|
| KMS Integration | Customer master keys (CMK) only — not default AWS-managed keys |
| SSL/TLS in Transit | All data movement is encrypted in transit |
| IAM | Fine-grained access control: who can create/edit/run recipes |
| CloudWatch | Log job runs, monitor performance |
| CloudTrail | Audit all API calls for compliance |

**■■ REMEMBER / WATCH OUT**

- DataBrew KMS only supports CUSTOMER MASTER KEYS — not the default AWS-managed KMS keys
- Output ALWAYS goes to S3 — DataBrew cannot write directly to databases
- DataBrew is different from Data Wrangler: DataBrew is standalone Glue service; Data Wrangler is inside SageMaker Studio

# 8. Amazon Athena — Advanced Features

## 8.1 Athena Fundamentals (Quick Review)

Athena is a **serverless, interactive SQL query service for data in S3** powered by Apache Presto. Data stays in S3 — no loading, no ETL required.

| Supported Format | Type | Performance |
|---|---|---|
| ORC | Columnar, splittable | ■■■ Best — 30-90% cost savings |
| Parquet | Columnar, splittable | ■■■ Best — 30-90% cost savings |
| Avro | Row-based, splittable | ■■ Good |
| JSON | Row-based, human-readable | ■ Baseline |
| CSV / TSV | Row-based, human-readable | ■ Baseline |
| Compressed (Snappy/Gzip/LZO/Zlib) | Applies to above formats | Better cost, sometimes less splittable |

### Cost Model:

- **$5 per TB scanned — you pay for the amount of data read, not compute time**
- Successful OR cancelled queries count — failed queries do NOT charge
- DDL (CREATE/ALTER/DROP) — FREE
- **Use ORC/Parquet to save 30-90% on query costs AND improve performance**

## 8.2 Athena Workgroups

Workgroups let you **organize and isolate** different teams, apps, or workloads within Athena:

- Each workgroup has its own: query history, data scan limits, IAM policies, encryption settings
- **Data limits: restrict how much data a workgroup can scan (cost control)**
- **Integrates with: IAM (access control), CloudWatch (monitoring), SNS (alerts)**
- Useful for: separating dev vs prod, billing by team, enforcing different security policies

## 8.3 Athena ACID Transactions (Apache Iceberg)

Athena supports **ACID transactions** via integration with **Apache Iceberg** table format:

| ACID Property | What Athena/Iceberg Provides |
|---|---|
| Atomicity | All changes in a transaction succeed or none do |
| Consistency | Database always in valid state |
| Isolation | Concurrent users can safely make row-level modifications |
| Durability | Committed data persists |

### How to Enable Iceberg in Athena:

```
CREATE TABLE my_table ( id bigint, name string ) WITH ( table_type = 'ICEBERG', location =
's3://my-bucket/my_table/' );
```

### Key Iceberg Features:

- **Time Travel: Recover recently deleted data with a SELECT statement — query past snapshots**
- **Row-level modifications: UPDATE, DELETE, MERGE INTO statements**
- **Schema evolution: Add/rename/drop columns without rewriting data**
- **Concurrent writes: Multiple users can write simultaneously without conflicts**
- **Compatible with: EMR, Spark, any service supporting Iceberg format**
- **No custom record locking needed — Iceberg handles this**

### Compaction (Performance Maintenance):

```
OPTIMIZE my_table REWRITE DATA USING BIN_PACK WHERE catalog = 'c1';
```

Run periodic compaction to maintain query performance as Iceberg creates many small files over time.

## 8.4 Athena Security

| Security Layer | Details |
|---|---|
| Access Control | IAM policies, S3 ACLs, S3 bucket policies |
| Managed Policies | AmazonAthenaFullAccess, AWSQuicksightAthenaAccess |
| Encryption at Rest | SSE-S3 (S3-managed), SSE-KMS, CSE-KMS |
| Encryption in Transit | TLS between Athena and S3 |
| Cross-Account Access | Configure via S3 bucket policies |
| Workgroup Encryption | Each workgroup can have independent encryption settings |

## 8.5 Athena Performance Optimization

| Optimization Technique | How / Why |
|---|---|
| Use columnar formats | ORC or Parquet — read only needed columns, not full rows. 30-90% cost savings |
| Use partitions | Filter on partition keys to skip large chunks of data |
| Fewer, larger files | Small number of large files >> large number of small files |
| Repair partitions | After adding partitions: MSCK REPAIR TABLE my_table |
| CTAS to convert formats | Use CREATE TABLE AS SELECT to convert CSV → Parquet in place |
| Compress data | Snappy compression with Parquet/ORC for best balance |

## 8.6 CREATE TABLE AS SELECT (CTAS)

CTAS is a powerful pattern to **convert data formats and create subsets** within Athena:

### Convert CSV to Parquet:

```
CREATE TABLE new_parquet_table WITH ( format = 'PARQUET', write_compression = 'SNAPPY' ) AS
SELECT * FROM old_csv_table;
```

**Convert to ORC with custom S3 location:**

```
CREATE TABLE my_orc_table WITH ( external_location = 's3://my-results/orc_data/', format =
'ORC' ) AS SELECT * FROM old_table;
```

- CTAS is the standard way to **migrate data to columnar format** within the Athena/S3 ecosystem
- Creates a new table — original table is preserved

## 8.7 Athena Fine-Grained Access to Glue Data Catalog

Control who can do what with tables and databases in the Glue Data Catalog via IAM:

| Operation to Control | IAM Actions Needed |
|---|---|
| Read database/table | glue:GetDatabase, glue:GetDatabases, glue:GetTable |
| Create database/table | glue:CreateDatabase, glue:CreateTable |
| Drop database | glue:DeleteDatabase, glue:GetDatabase |
| Drop table | glue:DeleteTable, glue:GetTable, glue:GetPartitions, glue:DeletePartition |
| ALTER / REPAIR TABLE | glue:UpdateTable, glue:BatchCreatePartition |
| SHOW DATABASES / TABLES | glue:GetDatabases, glue:GetTables |

**Important:** At minimum you must grant access to your **database AND the Glue Data Catalog in each region**. Forgetting the catalog-level permission is a common misconfiguration.

> ■ **EXAM TIP**
> Athena anti-patterns exam trick: If the question asks about 'highly formatted reports / dashboards' → QuickSight. 'ETL transforms on S3 data' → Glue. Athena is for ad-hoc SQL queries and analysis, NOT ETL or visualization.

# Quick Reference — Service Comparisons

## Data Preparation Tools Comparison

| Service | Who Uses It | Interface | Best For | Output Goes To |
|---|---|---|---|---|
| SageMaker Data Wrangler | Data Scientists | Visual GUI (Studio) | ML feature engineering, 300+ transforms | S3, Feature Store, Pipeline |
| AWS Glue DataBrew | Data Analysts | Visual GUI (standalone) | General data prep, PII handling, 250+ transforms | S3 only |
| AWS Glue Studio | Data Engineers | Visual DAG builder | Complex ETL pipelines, Kinesis/Kafka sources | S3, Glue Catalog |
| AWS Glue ETL Jobs | Data Engineers | Code (Python/Scala) | Custom Apache Spark transformations | S3, databases |
| Amazon Athena | Analysts, Scientists | SQL | Ad-hoc queries, format conversion (CTAS) | S3 (query results) |

## SageMaker Services — When to Use What

| Need | SageMaker Service |
|---|---|
| Label training data (humans) | Ground Truth |
| Fully managed labeling (AWS does everything) | Ground Truth Plus |
| Visual data preparation | Data Wrangler |
| Store and share ML features centrally | Feature Store |
| Monitor deployed model for drift | Model Monitor |
| Detect bias and explain predictions | Clarify |
| No-code ML for business analysts | Canvas |
| Processing/feature engineering jobs | Processing Jobs |
| Train with built-in algorithms | Training Jobs |
| Real-time predictions (single) | Persistent Endpoint |
| Batch predictions (whole dataset) | Batch Transform |
| Deploy to IoT/edge | Neo |

## Exam Keyword → Service Mapping

| Keyword / Phrase in Question | Answer Service |
|---|---|
| 'Consistent features between training and inference' | SageMaker Feature Store |

| Keyword / Phrase in Question | Answer Service |
|---|---|
| 'Share features across teams/models' | SageMaker Feature Store |
| 'Explain model predictions / feature importance' | SageMaker Clarify (SHAP) |
| 'Detect model bias before training' | SageMaker Clarify (pre-training metrics) |
| 'Monitor deployed model for data drift' | SageMaker Model Monitor |
| 'Visual no-code data prep in SageMaker' | SageMaker Data Wrangler |
| 'Visual data prep outside SageMaker' | AWS Glue DataBrew |
| 'PII handling, masking, hashing' | AWS Glue DataBrew |
| 'Serverless SQL on S3' | Amazon Athena |
| 'ETL pipelines, Spark transforms' | AWS Glue |
| 'Schema discovery, metadata catalog' | AWS Glue Crawler + Data Catalog |
| 'ACID transactions on S3 data' | Athena + Apache Iceberg |
| 'Time travel / recover deleted data' | Athena + Apache Iceberg |
| 'Human labeling of training data' | SageMaker Ground Truth / MTurk |
| 'Crowdsource labeling task' | Amazon Mechanical Turk |
| 'Active learning to reduce labeling cost' | SageMaker Ground Truth (70% savings) |

## Security Summary — All Services

| Service | Encryption at Rest | In Transit | Access Control | Special Notes |
|---|---|---|---|---|
| Feature Store | ■ KMS CMK | ■ TLS | IAM | PrivateLink supported |
| Data Wrangler | Via S3 | ■ TLS | IAM + Studio | — |
| Glue DataBrew | ■ KMS CMK only | ■ SSL | IAM + CloudTrail | CMK only — not AWS-managed keys |
| Athena | SSE-S3 / SSE-KMS / CSE-KMS | ■ TLS | IAM + ACL + bucket policy | Cross-account via S3 policy |
| Model Monitor | S3 encryption | ■ TLS | IAM | Metrics to CloudWatch |
| EMR Serverless | S3 encryption (SSE/CSE) | ■ TLS | IAM + Kerberos | Spark comms encrypted |

# ■ Final Exam Cheat Sheet — Part 2

| | |
|---|---|
| **SageMaker Training Data** | Always in Amazon S3. Training code in ECR. Model artifacts output to S3. |
| **Pipe Mode vs File Mode** | Pipe = stream from S3 (fast, large data, RecordIO). File = copy to disk (simple). |

| | |
|---|---|
| **Ground Truth 70%** | Active learning loop reduces labeling cost by UP TO 70%. |
| **Feature Store Online vs Offline** | Online = real-time inference (PutRecord/GetRecord API). Offline = training (S3 + Athena). |
| **Model Monitor 4 Types** | Data quality / Model quality / Bias drift / Feature attribution drift (NDCG score). |
| **Clarify = SHAP** | SageMaker Clarify uses SHAP for feature importance. Asymmetric SHAP for time series. |
| **Athena Pricing** | $5/TB scanned. ORC/Parquet saves 30-90%. Failed queries = free. |
| **Athena ACID = Iceberg** | Add table_type='ICEBERG' to CREATE TABLE. Enables time travel and row-level updates. |
| **Glue Crawler** | Scans S3/databases → populates Data Catalog → enables Athena/Redshift Spectrum/EMR. |
| **CTAS in Athena** | CREATE TABLE AS SELECT — converts data formats (CSV to Parquet) AND creates subsets. |
| **DataBrew KMS** | DataBrew only supports Customer Master Keys (CMK) — NOT default AWS-managed KMS keys. |
| **VPC Only Mode** | SageMaker Domain 'VPC Only' = all traffic through your private VPC, no internet. |
| **Batch Transform vs Endpoint** | Batch Transform = entire dataset at once. Endpoint = real-time single predictions. |
| **SageMaker Neo** | Deploy ML models to edge devices (IoT). Compiles model for target hardware. |
| **Data Wrangler Quick Model** | Train a fast model to VALIDATE your feature engineering — check feature importance. |
| **MTurk + Confidential Data** | NEVER use Mechanical Turk for confidential/proprietary data — workers are public. |
| **MSCK REPAIR TABLE** | Run after adding new S3 partitions to Athena/Glue to update the catalog. |